The Unit Circle and Exact Values

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exact values**

Write the exact value of each expression:

a $\sin\frac{\pi}{6} = \frac{1}{2}$ **b** $\cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$ **c** $\tan\frac{5\pi}{6} = -\frac{\sqrt{3}}{3}$ **d** $\sin\frac{3\pi}{2} = -1$

(a) $= \frac{1}{2}$; (b) $= -\frac{\sqrt{2}}{2}$; (c) $= -\frac{\sqrt{3}}{3}$; (d) $= -1$

2 **[Foundational]** • **Target: Points on the unit circle**

The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle. Find:

a $\sin\theta = \frac{4}{5}$ **b** $\cos\theta = -\frac{3}{5}$

c $\tan\theta = -\frac{4}{3}$ **d** the quadrant containing θ Quadrant II

(a) $= \frac{4}{5}$; (b) $= -\frac{3}{5}$; (c) $= -\frac{4}{3}$; (d) Quadrant II

3 **[Foundational]** • **Target: Points on the unit circle**

State the reference angle for each angle:

a $\frac{5\pi}{6} = \frac{\pi}{6}$ **b** $\frac{4\pi}{3} = \frac{\pi}{3}$ **c** $-\frac{\pi}{4} = \frac{\pi}{4}$ **d** $\frac{7\pi}{2} = \frac{\pi}{2}$

(a) $= \frac{\pi}{6}$; (b) $= \frac{\pi}{3}$; (c) $= \frac{\pi}{4}$; (d) $= \frac{\pi}{2}$

4 **[Foundational]** • **Target: Points on the unit circle**

Identify the quadrant(s) in which each pair of conditions holds:

a $\sin\theta < 0$ and $\cos\theta > 0$ Quadrant IV.

b $\tan\theta > 0$ and $\sin\theta < 0$ Quadrant III.

(a) Quadrant IV.; (b) Quadrant III.

5 **[Foundational]** • **Target: Points on the unit circle**

Given that $\cos\theta = -\frac{5}{13}$ and θ is in Quadrant III, find $\sin\theta$ and $\tan\theta$ exactly.

$\sin^2\theta = 1 - \frac{25}{169} = \frac{144}{169}$, and in QIII $\sin\theta < 0$: $\sin\theta = -\frac{12}{13}$. $\tan\theta = \frac{-12/13}{-5/13} = \frac{12}{5}$.

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6 [Foundational] • Target: Even/odd properties and periodicity

Use the fact that cosine is even and sine is odd to simplify each expression:

a $\sin(-\theta) + \sin\theta \ = 0$

b $\cos(-\theta) - \cos\theta \ = 0$

c $\frac{\sin(-\theta)}{\cos(-\theta)} \ = -\tan\theta$

(a) $= 0$; (b) $= 0$; (c) $= -\tan\theta$

7 [Foundational] • Target: Even/odd properties and periodicity

Use periodicity to find the exact value of each expression:

a $\sin\frac{13\pi}{6}$ $\frac{13\pi}{6} = 2\pi + \frac{\pi}{6}$: $= \frac{1}{2}$ **b** $\cos\frac{17\pi}{4}$ $\frac{17\pi}{4} = 4\pi + \frac{\pi}{4}$: $= \frac{\sqrt{2}}{2}$ **c** $\tan\frac{9\pi}{4}$ $\frac{9\pi}{4} = 2\pi + \frac{\pi}{4}$: $= 1$

(a) $\frac{13\pi}{6} = 2\pi + \frac{\pi}{6}$: $= \frac{1}{2}$; (b) $\frac{17\pi}{4} = 4\pi + \frac{\pi}{4}$: $= \frac{\sqrt{2}}{2}$; (c) $\frac{9\pi}{4} = 2\pi + \frac{\pi}{4}$: $= 1$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving basic trigonometric equations on the unit circle

Find all solutions to $\sin\theta = -\frac{1}{2}$ in the interval $[0, 2\pi)$.

Reference angle $\frac{\pi}{6}$, sine negative in QIII and QIV: $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$.

9 [Intermediate] • Target: Solving basic trigonometric equations on the unit circle

Find all solutions to $2\cos\theta + \sqrt{3} = 0$ in the interval $[0, 2\pi)$.

$\cos\theta = -\frac{\sqrt{3}}{2}$: reference angle $\frac{\pi}{6}$, cosine negative in QII and QIII: $\theta = \frac{5\pi}{6}, \frac{7\pi}{6}$.

10 [Intermediate] • Target: Building the unit circle from special triangles

Using the 30-60-90 triangle, derive the exact coordinates of the point on the unit circle at $\theta = \frac{\pi}{3}$.

A 30-60-90 triangle with hypotenuse 1 has legs $\frac{1}{2}$ and $\frac{\sqrt{3}}{2}$. At $\theta = \frac{\pi}{3}$ (60°), $x = \cos\frac{\pi}{3} = \frac{1}{2}$, $y = \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$. Point: $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$.

11 [Intermediate] • Target: Building the unit circle from special triangles

State the coordinates of all four points on the unit circle where the terminal side lies along an axis ($\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$).

$\theta = 0$: $(1, 0)$. $\theta = \frac{\pi}{2}$: $(0, 1)$. $\theta = \pi$: $(-1, 0)$. $\theta = \frac{3\pi}{2}$: $(0, -1)$.

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12 [Intermediate] • Target: Using the Pythagorean identity with unit circle points

A point on the unit circle has x-coordinate $\frac{7}{25}$ and lies in Quadrant IV. Find its y-coordinate using $x^2 + y^2 = 1$.

$y^2 = 1 - \frac{49}{625} = \frac{576}{625}$, and in QIV $y < 0$: $y = -\frac{24}{25}$.

13 [Intermediate] • Target: Using the Pythagorean identity with unit circle points

Given $\sin\theta = 0.6$ and θ in Quadrant II, find $\cos\theta$ and $\tan\theta$ exactly (using the fact $0.6 = \frac{3}{5}$).

$\cos^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$, and in QII $\cos\theta < 0$: $\cos\theta = -\frac{4}{5}$. $\tan\theta = \frac{3/5}{-4/5} = -\frac{3}{4}$.

14 [Intermediate] • Target: The unit circle picture

The unit circle $x^2 + y^2 = 1$ is shown. Use it to estimate $\cos\left(\frac{2\pi}{3}\right)$ and $\sin\left(\frac{2\pi}{3}\right)$ visually, then state the exact values.

The angle $\frac{2\pi}{3}$ $(120°)$ lies in QII, roughly two-thirds around from 0. Exact values: $\cos\frac{2\pi}{3} = -\frac{1}{2}$, $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: All six trig functions from a unit circle point • Cross-Lesson: 7.04 Applications: Growth and Decay

A point on the unit circle has coordinates $\left(-\frac{5}{13}, \frac{12}{13}\right)$. Find all six trigonometric function values for the corresponding angle θ.

$\sin\theta = \frac{12}{13}$, $\cos\theta = -\frac{5}{13}$, $\tan\theta = -\frac{12}{5}$, $\csc\theta = \frac{13}{12}$, $\sec\theta = -\frac{13}{5}$, $\cot\theta = -\frac{5}{12}$.

16 [Challenging] • Target: Solving for the angle given exact coordinates • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find all values of θ in $[0, 2\pi)$ for which the unit circle point is exactly $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.

This point lies in QIII where both coordinates are negative with reference angle $\frac{\pi}{4}$: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$ (only one solution in $[0, 2\pi)$ since the exact point pins down a single angle).

17 [Challenging] • Target: Solving basic trigonometric equations on the unit circle • Cross-Lesson: 8.02 Graphing Rational Functions

Consider a point P on the unit circle such that its coordinates are $(\cos\theta, \sin\theta)$. Let α be an angle in the first quadrant and β be an angle in the third quadrant, with $\cos\alpha = \frac{3}{5}$ and $\sin\beta = -\frac{4}{5}$. If the angle θ is such that $\theta = \alpha + \beta$, find the value of $\cos\theta$.

We are given $\cos\alpha = \frac{3}{5}$ and α is in the first quadrant. Using the identity $\sin^2\alpha + \cos^2\alpha = 1$, we find $\sin\alpha = \sqrt{1 - (\frac{3}{5})^2} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$ (since α is in the first quadrant, $\sin\alpha$ is positive). We are given $\sin\beta = -\frac{4}{5}$ and β is in the third quadrant. Using the identity $\sin^2\beta + \cos^2\beta = 1$, we find.

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18 [Challenging] • Target: Building the unit circle from special triangles • Cross-Lesson: 8.03 Introduction to Limits

On the unit circle, a point P is located at coordinates (x, y). If the angle formed by the positive x-axis and the line segment from the origin to P is θ, what is the value of x in terms of θ?

By definition, the coordinates of a point $P(x, y)$ on the unit circle corresponding to an angle θ (measured counterclockwise from the positive x-axis) are given by $(\cos\theta, \sin\theta)$. Therefore, the x-coordinate of the point P is $\cos\theta$.

19 [Challenging] • Target: Using the Pythagorean identity with unit circle points • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\sin\theta = \frac{3}{5}$ with θ terminating in Quadrant I, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Step 1: Compute $\cos\theta$ via Pythagorean identity: $\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$. Step 2: Compute $\tan\theta$: $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{3/5}{4/5} = \frac{3}{4}$. Step 3: Compute $\sin(2\theta)$ using the double-angle identity: $\sin(2\theta) = 2\sin\theta\cos\theta = 2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{24}{25}$.

20 [Challenging] • Target: The unit circle picture • Cross-Lesson: 9.02 Geometric Sequences and Series

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$.
$y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Coordinates on the Unit Circle

On the unit circle, what do the coordinates (x, y) of a point at angle θ represent?

a. $x = \cos\theta$ **and** $y = \sin\theta$ ✓

b. $x = \sin\theta$ and $y = \cos\theta$

c. $x = \tan\theta$ and $y = \cot\theta$

d. $x = r\cos\theta$ where $r > 1$

By definition of the unit circle ($r = 1$), the horizontal coordinate is $x = \cos\theta$ and the vertical coordinate is $y = \sin\theta$.

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22 [Foundational] • Target: Coordinates on the Unit Circle

Find the exact coordinates on the unit circle corresponding to $\theta = \frac{5\pi}{6}$.

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

In Quadrant II: $x = \cos(5\pi/6) = -\frac{\sqrt{3}}{2}$, $y = \sin(5\pi/6) = \frac{1}{2}$.

23 [Foundational] • Target: Coordinates on the Unit Circle

Find the exact coordinates on the unit circle for $\theta = \frac{4\pi}{3}$.

a. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ ✓

b. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

c. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

In Quadrant III: $x = \cos(4\pi/3) = -1/2$, $y = \sin(4\pi/3) = -\sqrt{3}/2$.

24 [Foundational] • Target: Reference Angles and Signs by Quadrant

Find the reference angle θ' for $\theta = 225^\circ$.

a. 45° ✓

b. 35°

c. 135°

d. 55°

In Quadrant III: $\theta' = \theta - 180^\circ = 225^\circ - 180^\circ = 45^\circ$.

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25 [Foundational] • Target: Reference Angles and Signs by Quadrant

In which quadrants is $\tan\theta$ positive?

a. Quadrants I and III ✓

b. Quadrants I and II

c. Quadrants I and IV

d. Quadrants II and IV

$\tan\theta = y/x$. It is positive where x and y have the same sign: Quadrant I (both positive) and Quadrant III (both negative).

26 [Foundational] • Target: Reference Angles and Signs by Quadrant

If $\cos\theta < 0$ and $\sin\theta > 0$, in which quadrant does the terminal ray of θ lie?

a. Quadrant II ✓

b. Quadrant I

c. Quadrant III

d. Quadrant IV

$x < 0$ and $y > 0$ defines Quadrant II.

27 [Foundational] • Target: Exact Values of the Six Trig Functions

Find the exact value of $\tan\left(\frac{2\pi}{3}\right)$.

a. $-\sqrt{3}$ ✓

b. $\sqrt{3}$

c. $-\frac{\sqrt{3}}{3}$

d. -1

Reference angle is $\pi/3$. In Quadrant II, tangent is negative: $-\tan(\pi/3) = -\sqrt{3}$.

28 [Foundational] • Target: Exact Values of the Six Trig Functions

Find the exact value of $\csc\left(\frac{7\pi}{4}\right)$.

a. $-\sqrt{2}$ ✓

b. $\sqrt{2}$

c. $-\frac{\sqrt{2}}{2}$

d. -2

$\sin(7\pi/4) = -\frac{\sqrt{2}}{2}$. Reciprocal gives $\csc(7\pi/4) = -\frac{2}{\sqrt{2}} = -\sqrt{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Exact Values of the Six Trig Functions

Evaluate $\cos(3\pi) - \sin\left(\frac{3\pi}{2}\right)$.

a. 0 ✓

b. -2

c. 2

d. 1

$\cos(3\pi) = \cos(\pi) = -1$. $\sin(3\pi/2) = -1$. Result $= -1 - (-1) = 0$.

30 [Intermediate] • Target: Quadrantal Angles and Negative Angles

Which of the following values is undefined?

a. $\tan(90^\circ)$ ✓

b. $\sin(90^\circ)$

c. $\cos(90^\circ)$

d. $\cot(45^\circ)$

$\tan(90^\circ) = \frac{\sin(90^\circ)}{\cos(90^\circ)} = \frac{1}{0}$, which is undefined.

31 [Intermediate] • Target: Quadrantal Angles and Negative Angles

Evaluate $\cos(-\pi/3)$.

a. $1/2$ ✓

b. $-1/2$

c. $\sqrt{3}/2$

d. $-\sqrt{3}/2$

Cosine is an even function: $\cos(-\theta) = \cos\theta$. Thus $\cos(-\pi/3) = \cos(\pi/3) = 1/2$.

32 [Intermediate] • Target: Quadrantal Angles and Negative Angles

Evaluate $\sin(-\pi/4)$.

a. $-\frac{\sqrt{2}}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. -1

d. 0

Sine is an odd function: $\sin(-\theta) = -\sin\theta$. Thus $\sin(-\pi/4) = -\frac{\sqrt{2}}{2}$.

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33 [Intermediate] • Target: Quadrantal Angles and Negative Angles

Consider the unit circle shown, where a point $P(x, y)$ lies on the circle in the first quadrant. A ray from the origin through P forms an angle θ with the positive x-axis. If the y-coordinate of P is $\frac{\sqrt{3}}{2}$ and $\cos\theta = \frac{1}{2}$, find the value of x.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{1}{2}$ ✓

c. 1

d. $\frac{\sqrt{2}}{2}$

The equation of a unit circle centered at the origin is $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $y = \frac{\sqrt{3}}{2}$. Substituting this into the equation of the unit circle: $x^2 + (\frac{\sqrt{3}}{2})^2 = 1$ $x^2 + \frac{3}{4} = 1$ $x^2 = 1 - \frac{3}{4}$ $x^2 = \frac{1}{4}$ Since point P lies in the first quadrant, its x-coordinate must be positive. Therefore, $x = \sqrt{\frac{1}{4}} = \frac{1}{2}$. Alternatively, we are given that the angle θ formed by the ray from the or.

34 [Intermediate] • Target: Exact values

Consider a unit circle centered at the origin. A point $P(x, y)$ lies on the unit circle in the second quadrant such that the angle formed by the positive x-axis and the line segment OP is θ. If $\sin(\theta) = \frac{3}{5}$ and the y-coordinate of P is $y = \frac{3}{5}$, determine the x-coordinate of P.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $-\frac{25}{16}$

The problem states that point $P(x, y)$ lies on the unit circle, meaning $x^2 + y^2 = 1$. We are given that the y-coordinate of P is $y = \frac{3}{5}$. Substituting this into the unit circle equation, we get $x^2 + (\frac{3}{5})^2 = 1$. This simplifies to $x^2 + \frac{9}{25} = 1$. Subtracting $\frac{9}{25}$ from both sides gives $x^2 = 1 - \frac{9}{25} = \frac{16}{25}$. Taking the square root of both sides, we get $x = \pm\frac{4}{5}$. The problem also states that point P is in the second quadrant, where the x.

35 [Intermediate] • Target: Points on the unit circle

Consider a point P on the unit circle in the second quadrant such that the y-coordinate of P is $\frac{3}{5}$. Let θ be the angle in standard position whose terminal side passes through P. Find the value of $\cos\theta$.

a. $-\frac{3}{5}$

b. $\frac{3}{5}$

c. $-\frac{4}{5}$ ✓

d. $\frac{4}{5}$

On the unit circle, any point P(x, y) satisfies the equation $x^2 + y^2 = 1$. We are given that the y-coordinate of point P is $\frac{3}{5}$ and P is in the second quadrant. For a point $P(\cos\theta, \sin\theta)$ on the unit circle, we have $x = \cos\theta$ and $y = \sin\theta$. Thus, we have $\sin\theta = \frac{3}{5}$. Substituting this into the unit circle equation: $\cos^2\theta + (\frac{3}{5})^2 = 1$. $\cos^2\theta + \frac{9}{25} = 1$. $\cos^2\theta = 1 - \frac{9}{25}$. $\cos^2\theta = \frac{16}{25}$..

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36 **[Intermediate]** • **Target: Even/odd properties and periodicity**

Consider a point $P(\cos\theta, \sin\theta)$ on the unit circle in the Cartesian plane. If the distance from the origin to the point $Q(x, y)$ is 5, and the angle between the positive x-axis and the line segment OQ is α, such that $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. If $x = \cos\alpha$ and $y = \sin\alpha$, find the maximum possible value of the product of the x-coordinates of P and Q, i.e., $\cos\theta \cdot \cos\alpha$.

a. 4/5

b. 3/5

c. **1** ✓

d. 5/3

The point P is on the unit circle, so its coordinates are $(\cos\theta, \sin\theta)$. We are given $\cos\theta = \frac{3}{5}$ and $\sin\theta = \frac{4}{5}$. The point Q has coordinates (x, y) and its distance from the origin is 5. The angle between the positive x-axis and OQ is α. Thus, $x = 5\cos\alpha$ and $y = 5\sin\alpha$. We are asked to find the maximum possible value of the product of the x-coordinates of P and Q, which is $\cos\theta \cdot x$. Substituting the given values, we h.

37 **[Intermediate]** • **Target: Solving basic trigonometric equations on the unit circle**

Consider a point P on the unit circle in standard position. If the angle θ in standard position has a terminal side passing through P such that $\cos(\theta) = \frac{1}{2}$ and $\sin(\theta) > 0$, find the value of $\sin(\theta)$.

a. $\frac{\sqrt{3}}{2}$ ✓

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. -1

For any point on the unit circle, the coordinates are $(\cos(\theta), \sin(\theta))$. The equation of the unit circle is $x^2 + y^2 = 1$. Substituting the coordinates, we get $\cos^2(\theta) + \sin^2(\theta) = 1$. We are given that $\cos(\theta) = \frac{1}{2}$. So, $\left(\frac{1}{2}\right)^2 + \sin^2(\theta) = 1$. This simplifies to $\frac{1}{4} + \sin^2(\theta) = 1$. Subtracting $\frac{1}{4}$ from both sides gives $\sin^2(\theta) = 1 - \frac{1}{4} = \frac{3}{4}$. Taking the square root of both sides, we get.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Building the unit circle from special triangles • Cross-Lesson: 8.03 Introduction to Limits

On the unit circle, a point $P(x, y)$ corresponds to an angle θ in standard position. If the angle $\theta = \frac{\pi}{3}$, what is the value of the y-coordinate of point P?

a. $\frac{1}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $\frac{\sqrt{3}}{2}$ ✓

d. 1

The coordinates of a point on the unit circle corresponding to an angle θ are given by $(\cos\theta, \sin\theta)$. In this case, $\theta = \frac{\pi}{3}$. Therefore, the y-coordinate of point P is $\sin\left(\frac{\pi}{3}\right)$. We know that $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$.

39 [Challenging] • Target: Using the Pythagorean identity with unit circle points • Cross-Lesson: 9.01 Arithmetic Sequences and Series

If $\cos\theta = \frac{60}{61}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{60}{61}$

b. $\frac{11}{61}$ ✓

c. $\frac{171}{610}$

d. $-\frac{11}{61}$

$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \frac{3600}{3721}} = \frac{11}{61}$.

40 [Challenging] • Target: The unit circle picture • Cross-Lesson: 9.02 Geometric Sequences and Series

If $\cos\theta = \frac{35}{37}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{157}{370}$

b. $-\frac{12}{37}$

c. $\frac{35}{37}$

d. $\frac{12}{37}$ ✓

$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \frac{1225}{1369}} = \frac{12}{37}$.

41 [Challenging] • Target: All six trig functions from a unit circle point • Cross-Lesson: 9.03 The Binomial Theorem

If $\cos\theta = \frac{45}{53}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{28}{53}$ ✓

b. $-\frac{28}{53}$

c. $\frac{333}{530}$

d. $\frac{45}{53}$

$\sin\theta = \sqrt{1 - \cos^2\theta} = \sqrt{1 - \frac{2025}{2809}} = \frac{28}{53}$.

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42 [Challenging] • Target: Solving for the angle given exact coordinates • Cross-Lesson: 1.01 Functions and Function Notation

If $\cos\theta = \frac{56}{65}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{79}{130}$

b. $\frac{33}{65}$ ✓

c. $-\frac{33}{65}$

d. $\frac{56}{65}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{3136}{4225}} = \frac{33}{65}$.

43 [Challenging] • Target: Coordinates on the Unit Circle • Cross-Lesson: 1.02 Domain and Range

If $\cos\theta = \frac{63}{65}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $-\frac{16}{65}$

b. $\frac{16}{65}$ ✓

c. $\frac{9}{26}$

d. $\frac{63}{65}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{3969}{4225}} = \frac{16}{65}$.

44 [Challenging] • Target: Reference Angles and Signs by Quadrant • Cross-Lesson: 1.03 Average Rate of Change

If $\cos\theta = \frac{55}{73}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{553}{730}$

b. $\frac{55}{73}$

c. $-\frac{48}{73}$

d. $\frac{48}{73}$ ✓

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{3025}{5329}} = \frac{48}{73}$.

45 [Challenging] • Target: Exact Values of the Six Trig Functions • Cross-Lesson: 2.01 Right-Triangle Trigonometry

If $\cos\theta = \frac{84}{85}$ and θ is in Quadrant I, find $\sin\theta$ using the Pythagorean identity.

a. $\frac{13}{85}$ ✓

b. $\frac{84}{85}$

c. $-\frac{13}{85}$

d. $\frac{43}{170}$

$\sin\theta = \sqrt{1-\cos^2\theta} = \sqrt{1-\frac{7056}{7225}} = \frac{13}{85}$.

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